



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
Configuration Number: **CN1301**
Material Type: **F3C 433/M65;38%;285AW;50"**
Lot Number: **92M0111528**
Order Number: **92M0111528**

Report Printed: 26-Mar-2026 4:29:26PM
Status: **PASS**
Approval: **TUCKERRR**

1. Tensile, SBS, and TG Panel

ASTM D8336 Nottingham Tack

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111528001/001	Tack Force (lbf)		0.930 - 1.035 - 1.100
	Tack Force Avg (lbf)		1.022

Dimensional Fiber Volume

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111528001/001	Laminate Density By Immersion Avg (g/cc)		1.5111
	Laminate Density By Immersion Std (g/cc)		0.0017
	Laminate Density By Immersion CoV		0.11
	Dimensional Fiber Volume (%)		53.88 - 53.97 - 54.29
	Dimensional Fiber Volume Avg (%)	50 - 56	54
	Dimensional Fiber Volume Std (%)		0.215
	Dimensional Fiber Volume CoV		0.40
	Void Volume (%)		1.3 - 1.2 - 1.2
	Void Volume Avg (%)	MAX= 2	1
	Void Volume Std (%)		0.070
	Void Volume CoV		5.79
92M0111528031/002	Laminate Density By Immersion Avg (g/cc)		1.5127
	Laminate Density By Immersion Std (g/cc)		0.0026
	Laminate Density By Immersion CoV		0.17
	Dimensional Fiber Volume (%)		53.23 - 54.77 - 53.96
	Dimensional Fiber Volume Avg (%)	50 - 56	54
	Dimensional Fiber Volume Std (%)		0.770
	Dimensional Fiber Volume CoV		1.43
	Void Volume (%)		1.0 - 1.2 - 1.1
	Void Volume Avg (%)	MAX= 2	1
	Void Volume Std (%)		0.105
	Void Volume CoV		9.94

Short Beam Shear Stress-Warp-Ambient

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111528001/001	SBS Strength (MPa)		79.9 - 85.6 - 86.4 - 87.2 - 89.1
	SBS Strength Avg (MPa)	MIN= 65	86



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
Configuration Number: **CN1301**
Material Type: **F3C 433/M65;38%;285AW;50"**
Lot Number: **92M0111528**
Order Number: **92M0111528**

Report Printed: 26-Mar-2026 4:29:26PM
Status: **PASS**
Approval: **TUCKERRR**

Short Beam Shear Stress-Warp-Ambient (Continued)

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111528001/001	SBS Strength Std (MPa)		3.44
	SBS Strength CoV		4.02
92M0111528031/002	SBS Strength (MPa)		83.0 - 78.9 - 91.9 - 83.3 - 85.7
	SBS Strength Avg (MPa)	MIN= 65	85
	SBS Strength Std (MPa)		4.77
	SBS Strength CoV		5.65

Short Beam Shear Stress-Warp-150C(302F)

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111528001/001	SBS Strength (MPa)		66.9 - 67.1 - 68.0 - 68.2 - 68.5
	SBS Strength Avg (MPa)	MIN= 50	68
	SBS Strength Std (MPa)		0.68
	SBS Strength CoV		1.00
92M0111528031/002	SBS Strength (MPa)		63.4 - 64.5 - 66.1 - 65.7 - 65.3
	SBS Strength Avg (MPa)	MIN= 50	65
	SBS Strength Std (MPa)		1.06
	SBS Strength CoV		1.63

Tg By DMA

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111528001/001	Tg by DMA in °C (° C)		262.0
	Tg by DMA in °C Avg (° C)	MIN= 225.0	262.0
92M0111528031/002	Tg by DMA in °C (° C)		267.0
	Tg by DMA in °C Avg (° C)	MIN= 225.0	267.0

Calculated Thickness/Ply

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111528001/001	Calculated Thickness/Ply Avg (mm)	0.28 - 0.32	0.29
	Calculated Thickness/Ply StDev (mm)		0.0051
	Calculated Thickness/Ply CoV		1.74
92M0111528031/002	Calculated Thickness/Ply Avg (mm)	0.28 - 0.32	0.30
	Calculated Thickness/Ply StDev (mm)		0.0051
	Calculated Thickness/Ply CoV		1.48



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
Configuration Number: **CN1301**
Material Type: **F3C 433/M65;38%;285AW;50"**
Lot Number: **92M0111528**
Order Number: **92M0111528**

Report Printed: 26-Mar-2026 4:29:26PM
Status: **PASS**
Approval: **TUCKERRR**

Tensile Stress, Modulus, and Elongation-Warp-Ambient

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111528001/001	Norm Tensile Strength (MPa)		941.8 - 900.0 - 927.5 - 1087.7 - 1098.0
	Norm Tensile Strength Avg (MPa)	MIN= 700	991
	Norm Tensile Strength Std (MPa)		94.25
	Norm Tensile Strength CoV		9.51
	Norm Modulus (GPa)		119.0 - 119.2 - 114.5 - 153.3 - 150.1
	Norm Modulus Avg (GPa)	MIN= 55	131
	Norm Modulus Std (GPa)		18.809
92M0111528031/002	Norm Modulus CoV		14.34
	Norm Tensile Strength (MPa)		874.6 - 985.7 - 969.2 - 957.6 - 985.5
	Norm Tensile Strength Avg (MPa)	MIN= 700	954
	Norm Tensile Strength Std (MPa)		46.22
	Norm Tensile Strength CoV		4.84
	Norm Modulus (GPa)		143.5 - 113.1 - 126.0 - 122.1 - 127.0
	Norm Modulus Avg (GPa)	MIN= 55	126
	Norm Modulus Std (GPa)		11.066
	Norm Modulus CoV		8.76

2. PHYSICAL PROPERTIES

Volatile content @100°C in pan

Roll	Test/Result:	Requirements:	Values:
92M0111528001	Volatile content @100°C in a Pan (%)		0.6 - 0.6
	Volatile content @100°C in a Pan Avg (%)	MAX= 1	1
92M0111528031	Volatile content @100°C in a Pan (%)		0.4 - 0.4
	Volatile content @100°C in a Pan Avg (%)	MAX= 1	0

3. PHYSICAL TACK

Tack

Roll	Test/Result:	Requirements:	Values:
92M0111528001	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111528003	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
Configuration Number: **CN1301**
Material Type: **F3C 433/M65;38%;285AW;50"**
Lot Number: **92M0111528**
Order Number: **92M0111528**

Report Printed: 26-Mar-2026 4:29:26PM
Status: **PASS**
Approval: **TUCKERRR**

Tack (Continued)

Roll	Test/Result:	Requirements:	Values:
92M0111528006	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111528008	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111528009	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111528010	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111528012	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111528014	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111528020	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111528022	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111528024	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111528026	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111528028	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111528030	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30

End of report